

Master 2 - DSC – MLDM

Text mining (Fiche 1)

Exercice 1

Q1 : In English, the fiftieth (50ème) most frequent term has a probability of appearance of 0.0018.

What is the constant value in the Zipf law for this language?

Q2: We suppose that the constant C of the Zipf law for a corpus of 10000 words is equal to 0.1. What is the rank of a word having 10 occurrences?

Exercice 2

Given a corpus composed of 3 documents D1, D2 et D3

D1	deux et deux quatre, quatre et quatre huit, huit et huit font seize Répétez dit le maître
D2	deux et deux quatre quatre et quatre huit huit et huit font seize Mais voilà un oiseau lyre qui passe dans le ciel
D3	Un enfant le voit un enfant entend un enfant appelle Sauve moi joue avec moi oiseau Alors un oiseau descend et joue avec un enfant

Question 1. Give the index size for this corpora

Question 2. Given the stop words list below, what is the index size after elimination of the stop words.

Stop word list: *et font le mais voilà un qui dans dit moi avec alors répétez passe voit entend appelle sauve joue descend*

Question 3. After elimination of the stop words, give a boolean representation and a tf representation of the documents.

Question 4 Compute the cosinus similarity between D1 and D2 with elimination of the stop words using the Boolean representation and the tf representation.

Question 5. What is the **inverse document frequency** of the word 'oiseau'?

Question 6. With the TF IDF weighting scheme ,

- What is the weight of the word 'oiseau' in the document D2?
- What is the weight of the word 'oiseau' in the document D3?

Exercice 3

We consider 6 documents d1,...d6 belonging to 3 categories (C1, C2, C3) and described according to the Boolean model with an index composed of 5 words (t1,...,t5)

- 1- Using the Khi2 statistic determine the most discriminant words for this collection
- 2- After elimination of the non discriminant words, propose a set of rules that can be used to identify the category of a new document.
- 3- Determine the category of the documents d7, d8 and d9 with these rules

	t1	t2	t3	t4	t5	
d1	1	1	1	0	1	c1
d2	1	0	1	0	0	c1
d3	0	0	1	1	1	c2
d4	0	0	1	1	0	c2
d5	0	0	0	1	1	c3
d6	0	0	0	1	0	c3
d7	1	0	0	0	1	?
d8	0	1	1	1	0	?
d9	0	1	0	0	0	?

Exercise 5 bis

Given a Term Document matrix X and its singular decomposition

$$X = \begin{pmatrix} 0 & -0.1 & 0 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad U = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$\Sigma = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 0.1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad V' = \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{pmatrix}$$

Question 1. Give an approximation of X and a representation for the documents and terms with k = 1.

Question 2. Give a representation for the documents and terms with k = 1

Exercise 7

We consider a corpus composed of four documents d1, d2, d3 and d4 represented with the boolean model. The Jaccard similarity between the documents is given below:

simJac	d1	d2	d3	d4
d1	1	0.33	0	0.25
d2	0.33	1	0.4	0.5
d3	0	0.4	1	0.66
d4	0.25	0.5	0.66	1

Use single linkage agglomerative clustering to group the 4 documents. Show the dendrogram. Do not forget to give the values for the aggregation measure.